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CRITICAL CARE

Management of PE

YG CG 1016 June 2022

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FIRST APPROVED: JUNE 2022 REVIEW BY: JUNE 2025

Key Points

- PE causes 10% of cardiac arrests but is rarely diagnosed or managed.
- In cardiac arrest, check for contra-indications then administer 50mg alteplace as IV bolus, followed by IV heparin. Continue CPR for 60-90 minutes. Use the ED LUCAS.
- In suspected PE with CV instability, administer 10mg alteplase bolus, followed by 90mg over 2 hours. Commence heparin bolus and infusion after initial alteplase bolus
- All patients with cardiac arrest or CV instability should be managed on ICU, except if precluded by frailty/co-morbidities.

Background

1. Pulmonary Embolus is common, with VTE being the third most common CV event after MI and stroke. Incidence is 39-115/100,000 and is 8 times more common in those over 80 compared to those in 50s.[1]
2. PE accounts for 10% of patients admitted with non-traumatic sudden death and 50% of those arriving with electromechanical dissociation or asystole on ECG.[2]
3. There are multiple guidelines on management of PE (British Thoracic Society, European Cardiology Society, ALS guidance), but there is limited evidence and guidance of PE in cardiac arrest. This is despite the fact that there are specific management points.
4. Thrombolysis carries a 9.9% rate of severe bleeding and a 1.7% rate of intracranial haemorrhage. However it significantly reduces mortality in high risk PE.
5. This guideline is designed to act as a summary to aid diagnosis and management for those with 'High Risk' suspected PE in cardiac arrest or who are cardiovascularly unstable.

Diagnosis

1. The diagnosis can be quantified with Well's score or the Geneva score. Irrespective of score used, the chances of PE are 10% in the low-probability category, 30% in the moderate-probability category, and 65% in the high probability.
2. Differential includes: cardiac tamponade, acute coronary syndrome, aortic dissection, acute valvular dysfunction, and hypovolaemia
3. In the presence of CV instability or cardiac arrest, formal diagnosis with CTPA can be omitted.
4. See appendix 1 for definition of CV instability, and appendix 2 for diagnostic algorithm.

PTO for management of Cardiac arrest and PE with CV instability

PE in Cardiac Arrest

1. Commence ALS.
2. Consider PE in 4H's and 4 T's in those with:
    a. Sudden PEA arrest and no ischaemic changes on ECG,
    b. pre-arrest symptoms of non-central chest pain, dyspnoea, pre-syncope, hypoxia or haemoptesis
    c. POCUS evidence of RV dilation or lower limb DVT.
3. Intubate early to optimise oxygenation and in expectation of prolonged resuscitation.
4. If PE is considered causative, check for Contra-indications to thrombolysis:

Absolute	**Relative**
History of haemorrhagic stroke or stroke of unknown origin	Transient ischaemic attack in previous 6 months
Ischaemic stroke in previous 6 months	Oral anticoagulation
Central nervous system neoplasm	Pregnancy or first post-partum week
Major trauma, surgery, or head injury in previous 3 weeks	Non-compressible puncture sites
Bleeding diathesis	Traumatic resuscitation
Active bleeding	Refractory hypertension (systolic BP >180 mmHg)
	Advanced liver disease
	Infective endocarditis
	Active peptic ulcer

4. Deliver thrombolysis: 50mg Alteplase in 50mls, Bolus via PVC or central.[3]
5. Immediately commence IV heparin as per BCU Heparing infusion chart.[1,2]
6. Continue ALS for 60-90 minutes. Consider LUCAS device to facilitate. Available from ED resus storeroom, but seek permission from ED Nurse or Consultant in Charge.

PE with CV instability: SBP <90 for 15 minute or need for vasopressors

1. Deliver 100% oxygen, targeting Sats >90%, as oxygen acts as a pulmonary vasodilator.[1]
2. Consider trial of HFNO2 or CPAP/NIV prior to RSI. There is a high risk of worsening haemodynamics with RSI in the presence of RV failure.[1]
3. For anaesthesia use drugs that minimise CV instability, such as ketamine or midazolam.[1]
4. Minimise PEEP, use 6ml/kg IBW, PAW <30.[1]
5. Consider 500ml fluid challenge. Avoid excessive volume, which will worsen RV function.[1]
6. Check for Contraindications to thrombolysis (see section 4, above). If contra-indicated, consider referral to Leicester for ECMO and Thrombectomy.
7. Administer thrombolysis: if weight >65Kg: Alteplase 10mg as IV bolus, followed by 90mg over 2 hours.[1] If Weight <65Kg, 1.5mg/kg over 2 hours.[3] If cardiac arrest is imminent 50mg bolus can be considered.[2]
8. Commence Noradrenaline for BP support. Noradrenaline does not cause pulmonary vasoconstriction, and will improve RV perfusion.[1]
9. If ongoing low cardiac output, commence dobutamine once noradrenaline established.[1]
10. Commence IV heparin infusion once thrombolysis infusion is complete.[1,2]
11. If fails to improve, consider referral to Leicester ECMO centre for ECMO + thrombectomy. RV at 36 hours for CV instability and evidence of ongoing RV failure.
12. All patients with cardiac arrest or CV instability should be cared for on ICU, unless frailty or co-morbidities precludes escalation.[1]

Appendix 1: Diagnostic definitions of CV instability, from ESC 2019.

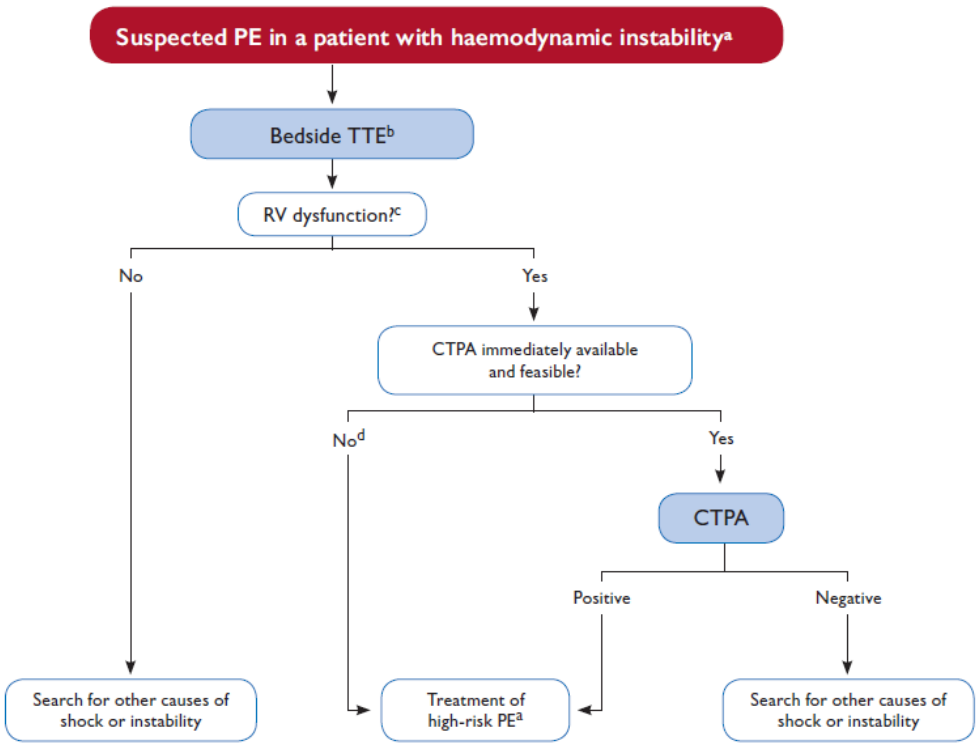
Table 4 **Definition of haemodynamic instability, which delineates acute high-risk pulmonary embolism (one of the following clinical manifestations at presentation)**

(1) Cardiac arrest	(2) Obstructive shock[68–70]	(3) Persistent hypotension
Need for cardiopulmonary resuscitation	Systolic BP < 90 mmHg or vasopressors required to achieve a BP ≥90 mmHg despite adequate filling status	Systolic BP < 90 mmHg or systolic BP drop ≥40 mmHg, lasting longer than 15 min and not caused by new-onset arrhythmia, hypovolaemia, or sepsis
	And	
	End-organ hypoperfusion (altered mental status; cold, clammy skin; oliguria/anuria; increased serum lactate)	

BP = blood pressure.

© ESC 2019

Appendix 2: Diagnostic protocol for PE with haemodynamic instability, from ESC 2019.



Add figures and diagrams in jpg. format

References

1) ESC 2019: 2019 ESC Guidelines for the diagnosis and management of acute pulmonaryembolism developed in collaboration with the European Respiratory Society (ERS). European Heart Journal (2020) 41, 543-603.
2) BTS 2003: British Thoracic Society guidelines for the management of suspected acute pulmonary embolism. *Thorax* 2003;**58:**470-483.
3) BCU PE Pathway V3, Betsi Sharepoint, accessed April 2022